

React HandsOn 9:-

THEORY

List the features of ES6

Some important ES6 features are:

- let and const keywords
- Arrow Functions (=>)
- Classes
- Class Inheritance
- Template Literals
- Destructuring
- Default Parameters
- Rest and Spread Operators
- Promises
- Modules (import/export)
- Map and Set Collections
- Enhanced Object Literals

Explain JavaScript let

let is used to declare variables whose values can be changed later.

Syntax

```
let age = 20;
```

```
age = 21;
```

Features

- Block scoped.
- Can be reassigned.
- Cannot be redeclared in the same scope.
- Prevents accidental variable redeclaration.

Identify the differences between var and let

var	let
Function scoped	Block scoped
Can be redeclared	Cannot be redeclared
Can be updated	Can be updated
Hoisted and initialized with undefined	Hoisted but not initialized (Temporal Dead Zone)
Older way of declaring variables	Introduced in ES6

Explain JavaScript const

const is used to declare variables whose values cannot be reassigned after initialization.

Syntax

```
const PI = 3.14;
```

Features

- Block scoped.
- Cannot be reassigned.
- Must be initialized during declaration.
- Used for constants.

Explain ES6 class fundamentals

A class is a blueprint for creating objects.

Syntax

```
class Student{  
  constructor(name, age){  
    this.name = name;  
    this.age = age;
```

```
}  
display(){  
    console.log(this.name + " " + this.age);  
}  
const s1 = new Student("Arpita",20);  
s1.display();
```

Output

Arpita 20

Features

- Uses class keyword.
- Uses constructor() for initialization.
- Supports methods.
- Creates reusable objects.

Explain ES6 class inheritance

Inheritance allows one class to acquire the properties and methods of another class using the extends keyword.

Example

```
class Animal{  
    speak(){  
        console.log("Animal speaks");  
    }  
}  
  
class Dog extends Animal{  
    bark(){  
        console.log("Dog barks");  
    }  
}
```

```
const d = new Dog();
```

```
d.speak();
```

```
d.bark();
```

Output

Animal speaks

Dog barks

Features

- Uses extends.
- Child class inherits parent class methods.
- Promotes code reuse.

Define ES6 Arrow Functions

Arrow functions provide a shorter syntax for writing functions.

Syntax

```
const functionName = () => {  
  // code  
};
```

Example

```
const add = (a,b) => a+b;
```

```
console.log(add(5,3));
```

Output

8

Identify Set() and Map()

Set

A Set is a collection of unique values.

Example

```
const numbers = new Set();
```

```
numbers.add(10);
```

```
numbers.add(20);  
numbers.add(10);  
console.log(numbers);
```

Output

Set(2) {10,20}

Features

- Stores only unique values.
- Duplicate values are ignored.
- Maintains insertion order.

Map

A Map stores key-value pairs.

Example

```
const student = new Map();  
student.set("Name","Arpita");  
student.set("Age",20);  
console.log(student.get("Name"));  
console.log(student.get("Age"));
```

Output

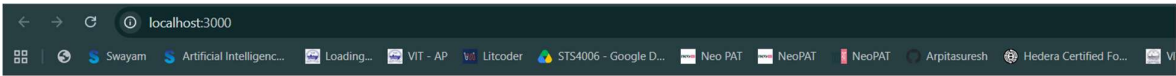
Arpita

20

Features

- Stores key-value pairs.
- Keys can be of any data type.
- Maintains insertion order.
- Provides methods like set(), get(), has(), and delete().

HANDS-ON LAB



List of Players

All Players

- Virat Kohli - 95
- Rohit Sharma - 82
- KL Rahul - 65
- Shubman Gill - 75
- Hardik Pandya - 60
- Ravindra Jadeja - 72
- R Ashwin - 55
- Mohammed Shami - 45
- Jasprit Bumrah - 35
- Kuldeep Yadav - 68
- Mohammed Siraj - 58

Players with Score Below 70

- KL Rahul - 65
- Hardik Pandya - 60
- R Ashwin - 55
- Mohammed Shami - 45
- Jasprit Bumrah - 35
- Kuldeep Yadav - 68
- Mohammed Siraj - 58